

5 (long)

(a) In the integrator circuit of Fig. 4(a) the op-amp can be considered ideal. At time $t < 0$, $v_{in} = 0$ and $v_{out} = 0$. Show that at time $t \geq 0$, v_{out} is given by

$$v_{out} = \beta \int_0^t v_{in} dt$$

and find an expression for the constant of proportionality β in terms of resistance R_1 and capacitance C . [6]

(b) The constant β is set to $\beta = -6.3 \times 10^3 \text{ s}^{-1}$. At time $t < 0$, $v_{in} = 0$ and $v_{out} = 0$. At time $t \geq 0$ a sinusoidal input v_{in} is applied to the circuit of Fig. 4(a). Sketch v_{out} as a function of t if v_{in} is given by:

(i) $v_{in} = 0.1 \sin(2\pi \times 1000t)$, that is, a sine wave of peak amplitude 0.1 V and frequency 1 kHz, [6]

(ii) $v_{in} = 0.1 \sin(2\pi \times 10t)$, that is, a sine wave of peak amplitude 0.1 V and frequency 10 Hz. [6]

(c) In the circuit of Fig. 4(b), the op-amp can be considered ideal. A resistor R_2 is in parallel with capacitor C .

(i) Derive an expression for the small-signal gain $|v_{out} / v_{in}|$ of the circuit in Fig 4(b). [6]

(ii) If the capacitance C in Fig. 4(b) is set to 3.18 nF, calculate the value of R_2 required to achieve a 3 dB cut-off frequency of 10 kHz. [6]

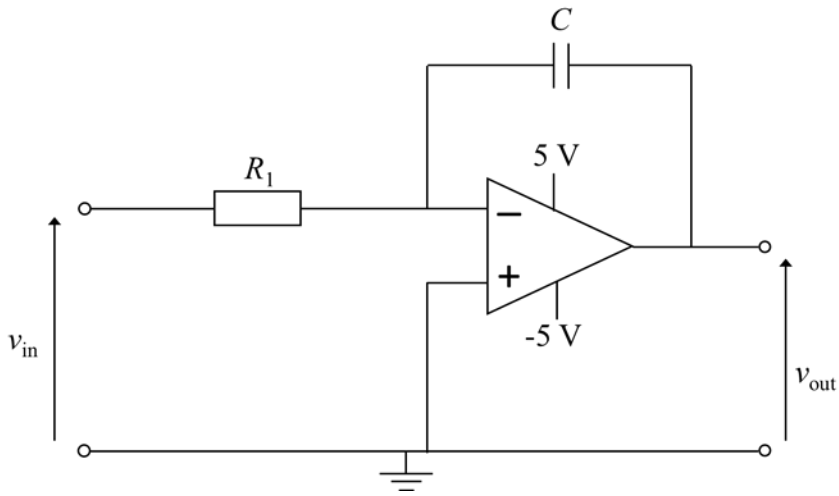


Fig. 4(a)

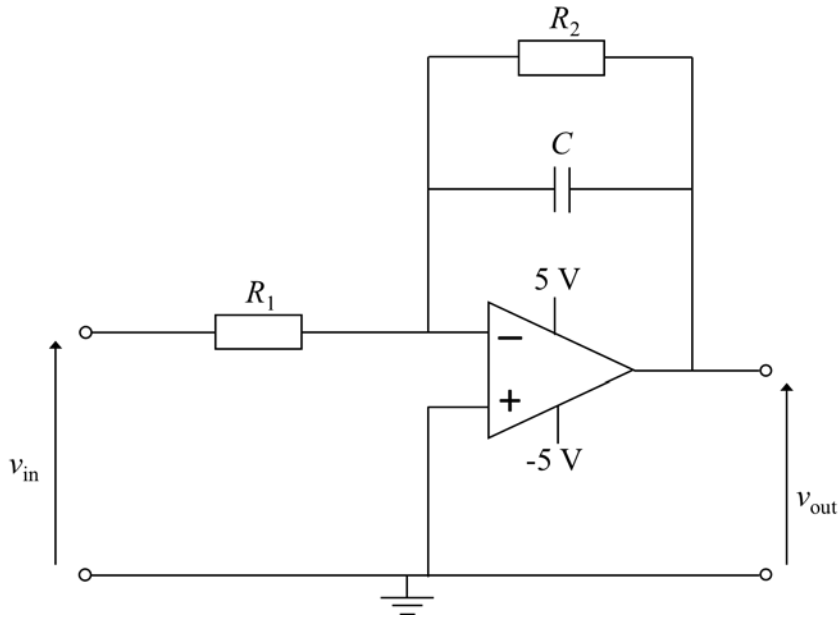


Fig. 4(b)